

Full-production spectral Galerkin convergence: adversarially repaired conditional theorem

TECT verification-first repository · claim A3-FULL-PRODUCTION-DISCRETIZATION-CONTINUUM

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1. Purpose and scope; repair provenance

This is the corrected integrated referee package for the quantitative P3 theorem. Version 2.1 is superseded because it used the continuum H^2 ball for the restarted exact-Galerkin trajectory without proving that the Fourier projection decreases the nonlinear energy. It also displayed selected upper bounds with inward decimal rounding, and its independent audit did not reconstruct the load-bearing constant chain. Those defects are not treated as cosmetic: the T6 conclusion below is re-enacted only after the separate Galerkin ball, outward rounding, and full-chain audit in this version.

The field is $u : \mathbb{T}^3 \rightarrow \mathbb{C}^3$, treated isometrically as six real components on the fixed torus with periods 16, 16, 16. The canonical P1 functional has the pinned positive regularisers, $\eta_{\text{shell}} = 0$, and density floor $\varepsilon_\rho = 10^{-12}$. The P2 gradient flow is

$$\partial_t u + Lu + N(u) = 0. \quad (1.1)$$

The single named hypothesis is

$$\text{A2-H3-CANONICAL-PRODUCTION-FUNCTIONAL :} \quad (H3)$$

the hash-pinned T5 P1 standalone reference functional is the canonical production functional. The T6 P2 well-posedness theorem is a hard dependency.

Corrected theorem. Assume (H3), let $R \in \{0.5, 1, 2\}$, and let $\|u_0\|_{H^2} \leq R$. For $\tau = 0.02$ and $T = 0.2$ the P2 solution satisfies

$$\sup_{\tau \leq t \leq T} \|u(t)\|_{H^6} \leq B_6(R, \tau, T) < \infty. \quad (1.2)$$

For every even Fourier grid size N , with P_N the real- L^2 Fourier projector and R_N^C the pinned collocation residual,

$$\sup_{\tau \leq t \leq T} \|P_N R(u(t)) - R_N^C(P_N u(t))\|_{L^2} \leq C(R, \tau, T) N^{-2}. \quad (1.3)$$

Let u_N solve the exact Galerkin equation

$$\partial_t u_N + Lu_N + P_N N(u_N) = 0, \quad u_N(\tau) = P_N u(\tau). \quad (1.4)$$

Then an explicit finite $E(R, \tau, T)$ satisfies

$$\sup_{\tau \leq t \leq T} \|u_N(t) - P_N u(t)\|_{L^2} \leq E(R, \tau, T) N^{-4}. \quad (1.5)$$

Equations (1.2)–(1.5) are a positive-time, restarted, exact-Galerkin conditional theorem. They do not assert an algebraic rate from $t = 0$ for merely H^2 data and do not identify finite oversampling with exact Galerkin for the rational Class-II term.

1 Continuum and discrete conventions

The real pairing and even Sobolev norms are

$$\langle v, w \rangle_{\mathbb{R}} = \operatorname{Re} \int_{\mathbb{T}^3} v^\dagger w, \quad \|v\|_{H^{2m}}^2 = |\mathbb{T}^3| \sum_k (1 + |k|^4)^m |\widehat{v}_k|^2. \quad (2.1)$$

The exact Galerkin residual is $P_N R(v_N)$. The executable collocation residual R_N^C is the real gradient of the periodic collocation energy. Pointwise nonlinear products alias, so these objects are never conflated.

The linear symbol contains the Brazovskii fourth-order scalar part, family mass, and internal lock. Direct spectral bounds give

$$c_L \|v\|_{H^2}^2 \leq \langle Lv, v \rangle \leq C_L \|v\|_{H^2}^2, \quad (2.2)$$

where $c_L = 0.2048572626782363$ and $C_L = 2.1566713536$ are derived from the pinned parameters.

2 Fourier, product, and Class-II constants

The max-shell count $24m^2 + 2$, an integral tail, and Hausdorff–Young give

$$\|v\|_\infty \leq E_2 \|v\|_{H^2}, \quad E_2 = 0.329926816704, \quad \|\partial_i v\|_{L^4} \leq E_{14} \|v\|_{H^2}, \quad E_{14} = 0.533391254984. \quad (3.1)$$

The Fourier-weight inequality and Young convolution give the deliberately enlarged algebra estimate

$$\|fg\|_{H^2} \leq A_2 \|f\|_{H^2} \|g\|_{H^2}, \quad A_2 = 8E_2 = 2.63941453363. \quad (3.2)$$

For $q_A = m_A/(\rho + \varepsilon_\rho)$ and $|u| \leq A$, inverse-density derivatives through order three obey

$$I_j = j!(2A + 2)^j \varepsilon_\rho^{-(j+1)}, \quad 0 \leq j \leq 3. \quad (3.3)$$

Leibniz convolution with the quadratic moment gives explicit envelopes Q_j for $D^j q_A$, and a second convolution gives metric envelopes G_j .

The contraction overcount is now explicit rather than merely declared. Write the Class-II energy density as $\frac{1}{2}B(u)[\nabla u, \nabla u]$. Its Euler term is

$$-\partial_i \{B(u) \partial_i u\} + \frac{1}{2} DB(u)^*[\nabla u, \nabla u]. \quad (3.4)$$

Expanding the divergence leaves one $DB[\nabla u, \nabla u]$ term and the half-weight term, hence a coordinate-free tensor factor $3/2$. The G_j already sum all three Pauli generators; the low-order formulas explicitly sum the three spatial directions; the H^2 formulas use (3.2); and the $\mathbb{C}^3 \simeq \mathbb{R}^6$ realification is isometric. Thus there is no unrecorded component multiplicity. The declared factor 108 has headroom

$$108/(3/2) = 72, \quad (3.5)$$

which the primary and independent audits both assert.

Consequently the pinned nonlinear map has computed majorants

$$\|N(u)\|_{L^2} \leq K_0(M), \quad \|N(u) - N(v)\|_{L^2} \leq L_0(M) \|u - v\|_{H^2}, \quad (3.6)$$

and analogous H^2 bounds $K_2(M_2, M_4)$ and $L_2(M_2, M_4)$. Every Q_j, G_j, K_s, L_s is persisted in the quantitative JSON with the density floor retained explicitly.

3 Continuum and Galerkin energy balls

The continuum energy-to-ball chain gives

$$M_2(R)^2 = \frac{2\{E_+(R) + C_Y|\mathbb{T}^3|\}}{c_L}. \quad (4.1)$$

For $R = 0.5, 1, 2$, respectively, the outward stored bounds for M_2 are 20.1609886852989, 20.3597732081305, and 21.2161594568883.

The crucial repair is that u_N is not assigned this same ball. Orthogonal projection gives only $\|P_N u(\tau)\|_{H^2} \leq M_2(R)$. We therefore apply the functional's direct upper estimate on that entire ball:

$$F(P_N u(\tau)) \leq E_+\{M_2(R)\}. \quad (4.2)$$

No assertion $F(P_N u) \leq F(u)$ is used. Exact-Galerkin energy dissipation and coercivity then give the independent uniform trajectory ball

$$M_{2,G}(R)^2 = \frac{2\{E_+(M_2(R)) + C_Y|\mathbb{T}^3|\}}{c_L}. \quad (4.3)$$

The three outward bounds are

$$M_{2,G} \leq 1457.64276910276, \quad 1500.96407138871, \quad 1697.46098407395. \quad (4.4)$$

Every evolution Lipschitz constant is evaluated on the common ball $M_* = \max(M_2, M_{2,G})$, not on the continuum ball alone.

4 Positive-time H^6 and residual bounds

For positive self-adjoint L , spectral calculus gives

$$\|L^\alpha e^{-tL}\| \leq (\alpha/(et))^\alpha. \quad (5.1)$$

Take $\alpha = 3/4$, Holder exponent $\theta = 1/4$, and $\delta = \tau/4$. The mild formula first bounds u in $D(L^{3/4})$. The endpoint cancellation

$$L \int_a^t S(t-s)f(s) ds = (I - S(t-a))f(t) + \int_a^t LS(t-s)\{f(s) - f(t)\} ds \quad (5.2)$$

changes the nonintegrable r^{-1} kernel to $r^{\theta-1}$ and gives an explicit H^4 bound. Repeating the argument on base space H^2 gives (1.2).

The Fourier projection and alias bounds are

$$\|(I - P_N)v\|_{H^2} \leq (L_{\max}/\pi)^4 N^{-4} \|v\|_{H^6}, \quad (5.3)$$

$$\|P_N(I_N - I)f\|_{L^2} \leq A_{\text{alias}} N^{-2} \|f\|_{H^2}, \quad A_{\text{alias}} = 167.383081344. \quad (5.4)$$

The repaired computation uses 90-digit decimal arithmetic and serialises upper bounds by directed ceiling rounding to 15 significant digits. For $R = 0.5, 1, 2$, the outward logarithmic bounds are

R	0.5	1	2
$\log_{10} B_6$	515.377687	515.751049	517.319858
$\log_{10} C$	576.354914	576.769779	578.512971

(5.5)

and (5.3)–(5.4) prove (1.3).

5 Repaired exact-Galerkin evolution estimate

Put $e = u_N - P_N u$. The continuum projection forcing is bounded on the continuum ball, while $N(u_N) - N(P_N u)$ is bounded on the common ball (4.4). Coercivity and Young's inequality give

$$\frac{d}{dt} \|e\|_{L^2} \leq a_G \|e\|_{L^2} + F_R N^{-4}, \quad a_G = \frac{L_0(M_*)^2}{2c_L}, \quad (6.1)$$

where $F_R = L_0(M_2)(L_{\max}/\pi)^4 B_6$. Since $e(\tau) = 0$,

$$\|e(t)\|_{L^2} \leq F_R \frac{e^{a_G(t-\tau)} - 1}{a_G} N^{-4}. \quad (6.2)$$

For $d = T - \tau = 0.18 \leq 1$, $(e^{a_G d} - 1)/a_G \leq d e^{a_G d} \leq e^{a_G d}$. Therefore the stored logarithmic enclosure is computed without a direction-sensitive subtraction:

$$\log_{10} E \leq \log_{10} F_R + \frac{a_G d}{\log 10}. \quad (6.3)$$

The outward stored values of this logarithm are $3.32936730872633 \cdot 10^{152}$, $5.97917890507636 \cdot 10^{152}$, and $6.99562180830297 \cdot 10^{153}$. Their extreme size makes the theorem non-practical but does not affect finiteness or the N^{-4} rate.

6 Independent checks and reproduction

The repaired primary audit passes 21/21. The non-importing independent audit passes 24/24 and reconstructs, from the pinned manifests, the continuum energy ball, separate Galerkin restart ball, coefficient and nonlinear envelopes, both endpoint bounds, residual constant, common-ball Lipschitz constant, growth rate, and Gronwall logarithm. It also verifies that projection-energy monotonicity is absent, checks the 72-fold contraction headroom, samples the semigroup and projection-tail inequalities, and retains the historical-solver exclusion. The integrated verifier passes 124/124, including the recorded CUDA evidence.

The one-command recorded-evidence reproduction is

```
python codes/foundations/a3_full_production_verify.py --reuse-recorded-audits
```

and must print `ASSERTS: 124/124` followed by `A3-FULL-PRODUCTION-VERIFY-PASS`. Omitting `-reuse-recorded-audits` re-executes all CPU audits in temporary directories.

7 Devil's-advocate review

1. **The continuum M_2 ball also bounds the Galerkin trajectory. UPHELD as unjustified in v2.1; repaired here.** Equations (4.2)–(4.4) derive a separate Galerkin ball without projection-energy monotonicity.
2. **A decimal approximation of an upper bound can be rounded to nearest. UPHELD as unsafe.** All stored upper bounds now use directed ceiling rounding; the graph lower bound uses directed floor rounding.
3. **The former 10/10 independent audit independently proved the constant chain. UPHELD as false.** It sampled only selected local ingredients. The repaired 24/24 audit reconstructs every load-bearing stage.
4. **The factor 108 hides unknown Class-II contractions. VALID WITH MITIGATION.** Equation (3.4) isolates the remaining factor 3/2 after generator, spatial, algebra, and realification conventions are accounted for; (3.5) records factor-72 headroom.

5. **Finite constants imply useful practical error bars. UPHELD as false.** The 10^{-12} floor makes the bounds astronomically large. The theorem establishes convergence only.
6. **H^2 data have the stated rate from $t = 0$. UPHELD as false.** Smoothing and comparison are used only after the positive restart time τ .
7. **A finite oversampling rule is exact for the rational Class-II residual. UPHELD as false.** Only mathematical projection $P_N N(u_N)$ is called exact Galerkin.
8. **T6 silently inherits a T5 functional. VALID WITH MITIGATION.** The exact identification is the named hypothesis (H3), satisfying dependency monotonicity.

8 Falsifier, tier review, and no-overclaim boundary

The theorem is falsified by source-hash drift; failure of the 124-assertion verifier or replacement bundle; an incorrect Fourier shell, endpoint, coefficient, energy-ball, directed-rounding, or alias estimate; a counterexample to (1.2)–(1.5); or failure of exact-Galerkin energy dissipation. Such a failure reopens A3-FULL-DISCRETIZATION-CLOSURE.

The original tier transition is T4 to T6 because this is a complete conditional theorem with one named sub-T6 hypothesis, a self-contained proof, explicit finite constants, independent reconstruction, and a PUBLISHED reproduction bundle. The adversarial finding temporarily challenged that justification; this v2.2 package re-enacts T6 rather than pretending the v2.1 argument was sound. T7 remains prohibited by the named hypothesis and absence of public external reproduction.

This theorem does not cover $t = 0$ algebraic rates for merely H^2 data, $\eta_{\text{shell}} \neq 0$, removed regularisers, infinite volume, finite-rule oversampling exactness, practical-grid error bars, historical Sector-B trajectories, full-spectrum Hessian convergence, minimizer or BCC selection, T7, or physical-domain closure.

Result footer

Result ID:	A3-FULL-PRODUCTION-DISCRETIZATION-CONTINUUM
Precise statement:	Explicit B6 and residual C on $[\tau, T]$, plus N^{-4} convergence of exact Galerkin restarted at τ .
Repair:	Separate Galerkin energy ball; common-ball Lip_0 ; outward rounding; full-chain independent audit.
Scope:	T6 CONDITIONAL-THEOREM; fixed T3, canonical P_1/P_2 , $\eta_{\text{shell}}=0$, $R=0.5/1/2$, $\tau=0.02$, $T=0.2$.
Named hypothesis:	A2-H3-CANONICAL-PRODUCTION-FUNCTIONAL.
Evidence grade:	ANALYTIC + EXECUTED + CONDITIONAL.
Reproduction command:	python codes/foundations/ a3_full_production_verify.py --reuse-recorded-audits
Expected output:	ASSERTS: 124/124; A3-FULL-PRODUCTION-VERIFY-PASS.
Falsification gate:	Failed hash, energy ball, coefficient/endpoint/ Fourier estimate, exact-Galerkin identity, or audit.
Tier before / after:	T6 challenged / T6 re-enacted; original T4-to-T6 promotion is justified only by this repaired package.
No-overclaim statement:	Not a practical solver error bar, $t=0$ H^2 rate, finite-oversampling exactness, Sector-B certificate, T7, minimizer, or BCC result.
Next required action:	Preserve as the repaired P3 baseline; sharper constants or executable solver bridges are separate.